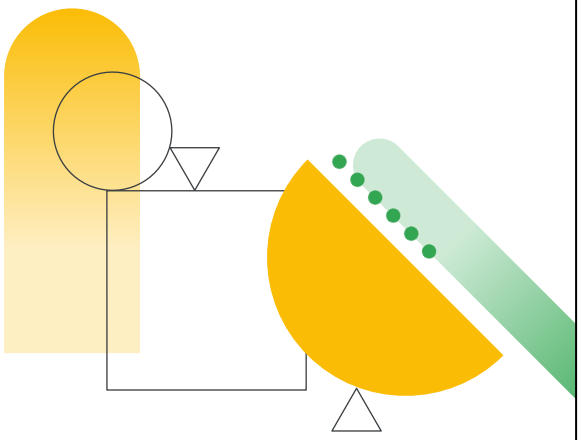
Google Cloud

DevOps Automation



This module introduces DevOps automation, a key factor in achieving consistency, reliability, and speed of deployment.

Proprietary + Confidential

Learning objectives

01 Automate service deployment using CI/CD pipelines.

02 Leverage Cloud Source Repositories for source and version control.

03 Automate builds with Cloud Build and build triggers.

04 Manage container images with Artifact Registry.

05 Investigate infrastructure with code using Terraform.

Google Cloud

Specifically, we will talk about the services that support continuous integration and continuous delivery practices, as part of a DevOps way of working.

With DevOps and microservices, automated pipelines for integrating, delivering, and potentially deploying code are required. These pipelines ideally run on on-demand provisioned resources. This module introduces the Google Cloud tools for developing code and creating automated delivery pipelines that are provisioned on demand.

We will talk about using Cloud Source Repositories for source and version control, Cloud Build, including build triggers, for automating builds, and managing containers with Artifact Registry.

We'll finish by reviewing infrastructure as code tools like Terraform.

Proprietary + Confidential

01

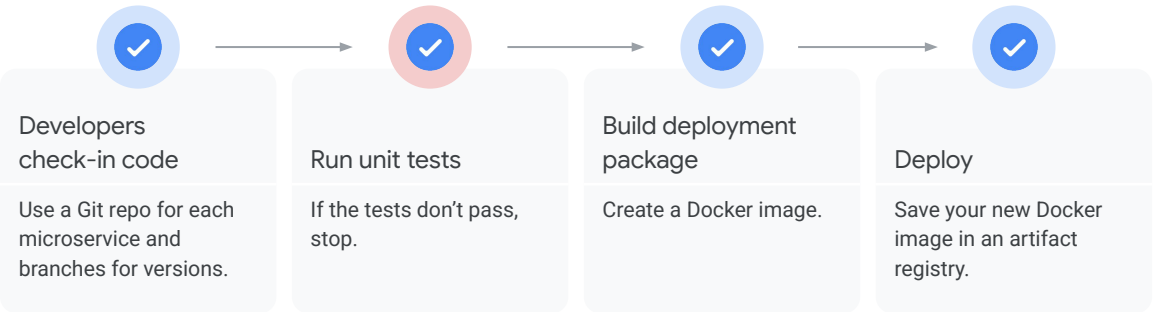
Continuous Integration Pipelines

Google Cloud

Let’s begin by talking about continuous integration pipelines.

Proprietary + Confidential

Continuous integration pipelines automate building applications



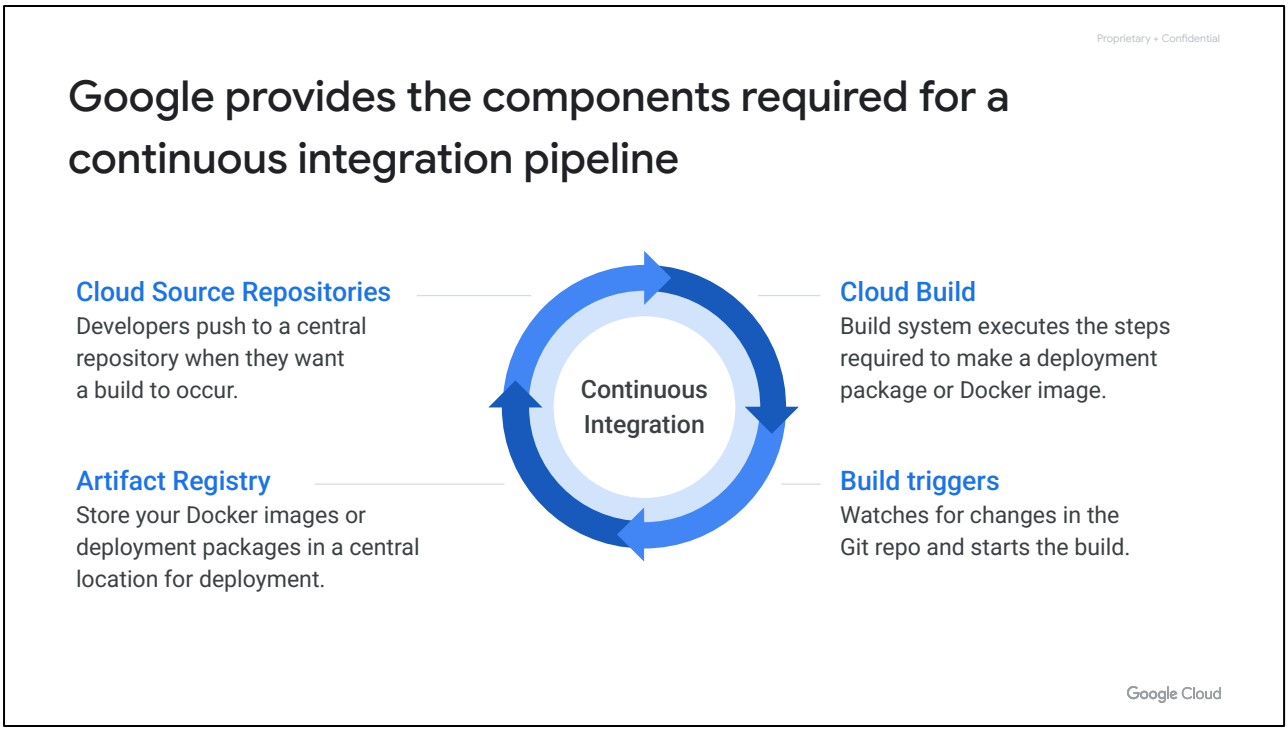
Google Cloud

Continuous integration pipelines automate building applications. This graphic shows a very simplistic view of a pipeline, which would be customized to meet your requirements.

The process starts by checking code into a repository where all the unit tests are run. On successful passing of these tests, a deployment package is built as a Docker image. This image is then saved in an artifact registry from where it can be deployed.

Each microservice should have its own repository.

Typical extra steps include linting of code/quality analysis by tools such as SonarQube, integration tests, generating test reports, and image scanning.



Google Cloud provides the components required to build a continuous integration pipeline. Let's go through each of those.

- The **Cloud Source Repositories** service provides private Git repositories hosted on Google Cloud. These repositories let you develop and deploy an app or service in a space that provides collaboration and version control for your code. Cloud Source Repositories are integrated with Google Cloud, so it provides a seamless developer experience.

- **Cloud Build** executes your builds on Google Cloud infrastructure. It can import source code from Cloud Storage, Cloud Source Repositories, GitHub, or Bitbucket, execute a build to your specifications, and produce artifacts such as Docker containers or Java archives. Cloud Build executes your build as a series of build steps, where each build step is run in a Docker container. A build step can do anything that can be done from a container, irrespective of the environment. There are standard steps, or you can define your own steps.

- A **Cloud Build trigger** automatically starts a build whenever you make changes to your source code. You can configure the trigger to build your code on any changes to the source repository or only changes that match certain criteria.

- **Artifact Registry** is a single place for your team to manage Docker images or deployment packages, perform vulnerability analysis, and decide who can

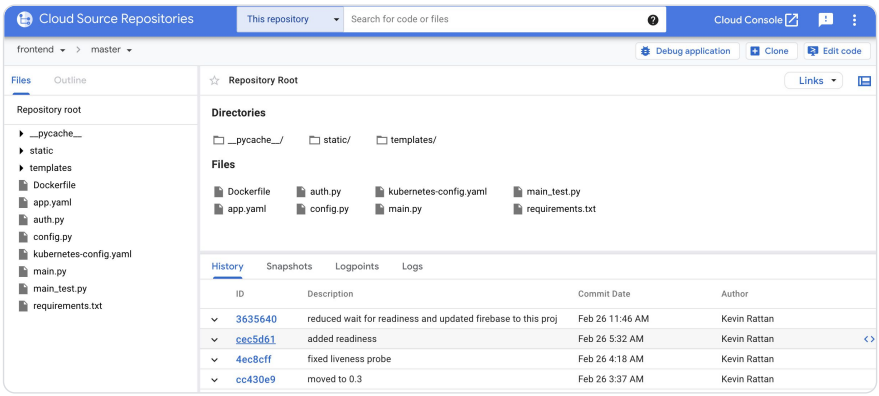
- access what with fine-grained access control.

Let’s go through each of these services in more detail.

Proprietary + Confidential

Cloud Source Repositories provides managed Git repositories

Control access to your repos using IAM within your Google Cloud projects.



Google Cloud

Cloud Source Repositories provide managed Git repositories.

You can use IAM to add team members to your project and grant them permissions to create, view, and update repositories.

Repositories can be configured to publish messages to a specific Pub/Sub topic. Messages can be published when a user creates or deletes a repository or pushes a commit.

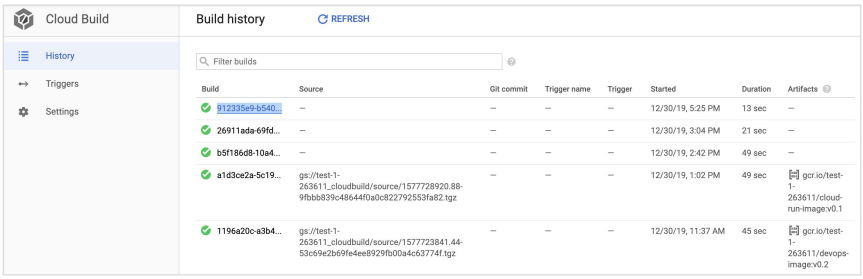
Some other features of Cloud Source Repositories include the ability to use audit logging to provide insights into what actions were performed where and when, and direct deployment to App Engine. It's also possible to connect an existing GitHub or Bitbucket repository to Cloud Source Repositories. Connected repositories are synchronized with Cloud Source Repositories automatically.

Proprietary + Confidential

Cloud Build lets you build software quickly across all languages

- Google-hosted Docker build service
  - Alternative to using Docker build command
- Use the CLI to submit a build

```
gcloud builds submit --tag gcr.io/your-project-id/image-name
```



Google Cloud

Developers gain complete control over defining workflows for building, testing, and deploying across multiple environments including VMs, serverless, and Kubernetes.

There is no more need to provision or maintain build environments: all is handled by Cloud Build. You write a build config to provide instructions to Cloud Build on what tasks to perform. These are defined as a series of steps. Each step is executed by a cloud builder. Cloud builders are containers with common languages and tools installed in them.

Builders can be configured to fetch dependencies, run unit tests, static analyses and integration tests, and create artifacts with build tools such as Docker, Gradle, Maven, Bazel, and Gulp.

Cloud Build executes the build steps you define. Executing build steps is similar to executing commands in a script.

You can either use the build steps provided by Cloud Build and the Cloud Build community or write your own custom build steps.

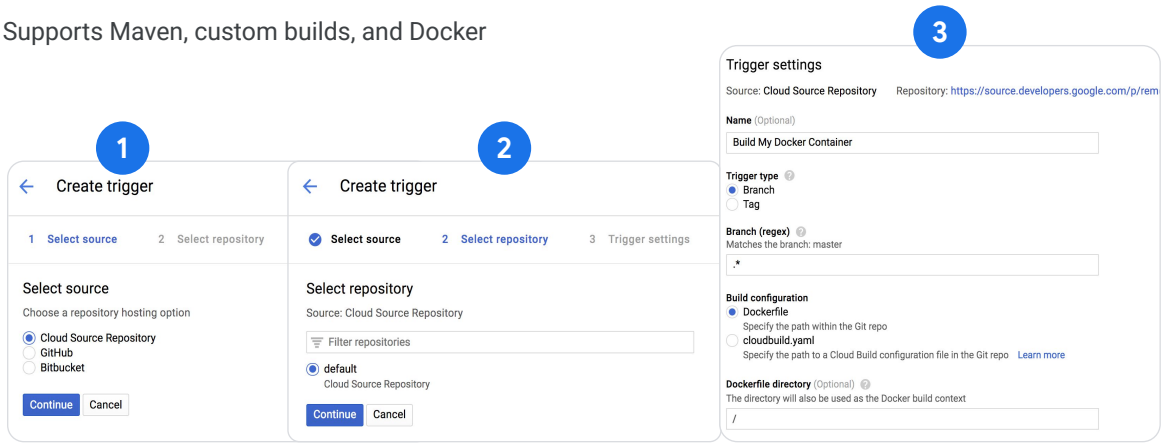
A GitHub link to a list of available cloud builders can be found in the course resources for this module.

[https://github.com/GoogleCloudPlatform/cloud-builders]

Proprietary + Confidential

Build triggers watch a repository and build a container whenever code is pushed

Supports Maven, custom builds, and Docker



Google Cloud

Build triggers watch a repository and build a container whenever code is pushed. Google's build triggers support Maven, custom builds, and Docker.

A Cloud Build trigger automatically starts a build whenever a change is made to source code. It can be set to start a build on commits to a particular branch or on commits that contain a particular tag. You can specify a regular expression to match the branch or tag value. A GitHub link to the syntax for the regular expression is available in this module's course resources.

The build configuration can be specified either in a Dockerfile or a Cloud Build file. The configuration required is shown on this slide.

First, a source is selected. This can be Cloud Source Repositories, Github, or Bitbucket. In the next stage, a source repository is selected, followed by the trigger settings. The trigger settings include information like the branch or tag to use for the trigger, and build configuration, for example the Dockerfile or Cloud Build file.

[Syntax: https://github.com/google/re2/wiki/Syntax]

Proprietary + Confidential

Artifact Registry is a universal package manager for build artifacts and dependencies

- Artifact Registry can store Docker and OCI container images in a Docker repository.
- Images built using Cloud Build are automatically saved in Artifact Registry.
  - Tag images with the prefix **`docker tag SOURCE-IMAGE LOCATION-docker.pkg.dev/PROJECT-ID/REPOSITORY/IMAGE:TAG`**
- Can use Docker push and pull commands with Artifact Registry.
  - `docker push LOCATION-docker.pkg.dev/PROJECT-ID/REPOSITORY/IMAGE:TAG`
  - `docker pull LOCATION-docker.pkg.dev/PROJECT-ID/REPOSITORY/IMAGE:TAG`

Google Cloud

Artifact Registry is a universal packet manager for build artifacts and dependencies. Artifact Registry can store Docker and OCI container images in a Docker registry.

Artifact Registry integrates with Google Cloud CI/CD services or your existing CI/CD tools. You can store artifacts from Cloud Build and deploy artifacts to Google Cloud runtimes, including Google Kubernetes Engine, Cloud Run, Compute Engine, and App Engine flexible environment. Identity and Access Management provides consistent credentials and access control.

Artifact Registry implements a Docker protocol so that you can push and pull images directly with Docker clients, including the Docker command-line tool.

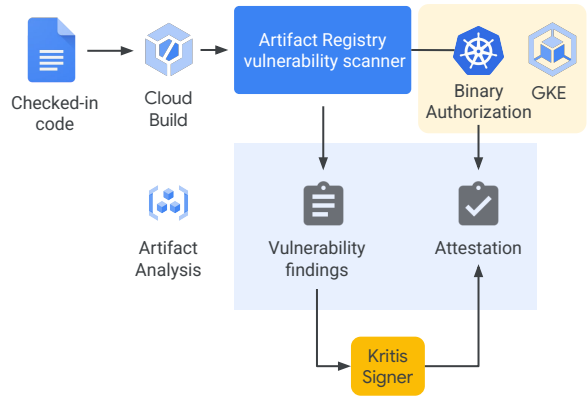
Google Cloud services that typically integrate with Artifact Registry, such as Cloud Build and Google Kubernetes Engine, are configured by default with permissions to access repositories in the same project and do not require a separate client.

Artifact Analysis is a family of services that provide software composition analysis, metadata storage and retrieval. Its detection points are built into a number of Google Cloud products such as Artifact Registry and Google Kubernetes Engine for easy and quick enablement. The service works with both Google Cloud's first-party products and also allows you to store information from third-party sources. The scanning services leverage a common vulnerability store for matching files against known vulnerabilities.

Proprietary + Confidential

Binary authorization allows you to enforce deploying only trusted containers into GKE

- Enable binary authorization on GKE cluster.
- Add a policy that requires signed images.
- When an image is built by Cloud Build an "attestor" verifies that it was from a trusted repository (Source Repositories, for example).
- Artifact Registry includes a vulnerability scanner that scans containers.



Google Cloud

Now binary authorization allows you to enforce deployment of only trusted containers into GKE. Binary authorization is a Google Cloud service and is based on the Kritis specification.

For this to work, you must enable binary authorization on your GKE cluster where your deployment will be made. A policy is required to sign the images. When an image is built by Cloud Build, an attestor verifies that it was from a trusted repository; for example, Source Repositories. Artifact Registry includes a vulnerability scanner that scans containers. A typical workflow is shown in the diagram.

Checking of code triggers a Cloud Build. As part of the build, Artifact Registry will perform a vulnerability scan when a new image is uploaded. The scanner publishes messages to Pub/Sub. The Kritis Signer listens to Pub/Sub notifications from an artifact registry vulnerability scanner and makes an attestation if the image scanning passed the vulnerability scan. Google Cloud binary authorization service then enforces the policy requiring attestations by the Kritis signer before a container image can be deployed.

More details can be found at:

https://cloud.google.com/binary-authorization/docs/cloud-build

https://cloud.google.com/binary-authorization/docs/vulnerability-scanning

Proprietary + Confidential

02

Infrastructure as Code

Google Cloud

Let’s now move on to consider infrastructure as code.

Proprietary + Confidential

Moving to the cloud requires a mindset change

On-Premises

- Buy machines.
- Keep machines running for years.
- Prefer fewer big machines.
- Machines are capital expenditures.

Cloud

- Rent machines.
- Turn machines off as soon as possible.
- Prefer lots of small machines.
- Machines are monthly expenses.

Google Cloud

Moving to the cloud requires a mindset change. The on-demand, pay-per-use model of cloud computing is a different model to traditional on-premises infrastructure provisioning. A typical on-premises model would be to buy machines and keep these running continuously. The compute infrastructure is typically built from fewer, larger machines. From an accounting view, the machines are capital expenditure that depreciates over time.

When using the cloud, resources are rented instead of purchased, and as a result we want to turn the machines off as soon as they are not required to save on costs. The approach is to typically have lots of smaller machines—scale out instead of scale up—and to expect and engineer for failure. From an accounting view, the machines are a monthly operating expense.

Proprietary + Confidential

In the cloud, all infrastructure needs to be disposable

- Don't fix broken machines.
- Don't install patches.
- Don't upgrade machines.
- If you need to fix a machine, delete it and re-create a new one.

- To make infrastructure disposable, automate everything with code:
  - Can automate using scripts.
  - Can use declarative tools to define infrastructure.

Google Cloud

In other words, in the cloud, all infrastructure needs to be disposable. The key to this is infrastructure as code (IaC), which allows for the provisioning, configuration, and deployment activities to be automated.

Having the process automated minimizes risks, eliminates manual mistakes, and supports repeatable deployments and scale and speed. Deploying one or one hundred machines is the same effort. The automation can be achieved using scripts or declarative tools such as Terraform which we will discuss later.

It is really important that no time is spent trying to fix broken machines or installing patches or upgrades. These will lead to problems recreating the environments at a later date. If a machine requires maintenance, remove it and create a new one instead.

Costs can be reduced by provisioning ephemeral environments, such as test environments that replicate the production environment.

Proprietary + Confidential

Infrastructure as code (IaC) allows for the quick provisioning and removing of infrastructures

- Build an infrastructure when needed.
- Destroy the infrastructure when not in use.
- Create identical infrastructures for dev, test, and prod.
- Can be part of a CI/CD pipeline.
- Templates are the building blocks for disaster recovery procedures.
- Manage resource dependencies and complexity.
- Google Cloud supports many IaC tools.

Google Cloud

Terraform is one of the tools used for Infrastructure as Code or IaC. Before we dive into understanding Terraform, let's look at what infrastructure as code does. In essence, infrastructure as code allows for the quick provisioning and removing of infrastructures.

The on-demand provisioning of a deployment is extremely powerful. This can be integrated into a continuous integration pipeline that smoothes the path to continuous deployment.

Automated infrastructure provisioning means that the infrastructure can be provisioned on demand, and the deployment complexity is managed in code. This provides the flexibility to change infrastructure as requirements change. And all the changes are in one place. Infrastructure for environments such as development and test can now easily replicate production and can be deleted immediately when not in use. All because of infrastructure as code.

Several tools can be used for IaC. Google Cloud supports Terraform, where deployments are described in a file known as a configuration. This details all the resources that should be provisioned. Configurations can be modularized using templates, which allows the abstraction of resources into reusable components across deployments.

In addition to Terraform, Google Cloud also provides support for other IaC tools, including:

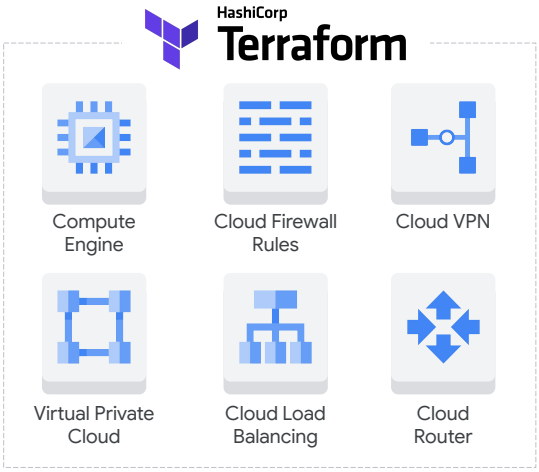
- Chef,
- Puppet,
- Ansible, and
- Packer

In this course however, we will focus on Terraform.

Proprietary + Confidential

Terraform is an infrastructure automation tool

- Repeatable deployment process
- Declarative language
- Focus on the application
- Parallel deployment
- Template-driven



Google Cloud

Terraform lets you provision Google Cloud resources—such as virtual machines, containers, storage, and networking—with declarative configuration files. You just specify all the resources needed for your application in a declarative format and deploy your configuration. HashiCorp Configuration Language (HCL) allows for concise descriptions of resources using blocks, arguments, and expressions.

This deployment can be repeated over and over with consistent results, and you can delete an entire deployment with one command or click. The benefit of a declarative approach is that it allows you to specify what the configuration should be and let the system figure out the steps to take.

Instead of deploying each resource separately, you specify the set of resources that compose the application or service, which allows you to focus on the application. Unlike Cloud Shell, Terraform will deploy resources in parallel.

Terraform uses the underlying APIs of each Google Cloud service to deploy your resources. This enables you to deploy almost everything we have seen so far, from instances, instance templates, and groups, to VPC networks, firewall rules, VPN tunnels, Cloud Routers, and load balancers.

For a full list of supported resource types, see the documentation at Using Terraform with Google Cloud.

Proprietary + Confidential

Terraform language

- Terraform language is the interface to declare resources.
- Resources are infrastructure objects.
- The configuration file guides the management of the resource.

```
resource "google_compute_network" "default" {
  name = "${var.network_name}"
  auto_create_subnetworks = false
}

<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> # Argument
}
```

Google Cloud

The Terraform language is the user interface to declare resources. Resources are infrastructure objects such as Compute Engine virtual machines, storage buckets, containers, or networks. A Terraform configuration is a complete document in the Terraform language that tells Terraform how to manage a given collection of infrastructure. A configuration can consist of multiple files and directories.

The syntax of the Terraform language includes:

- Blocks that represent objects and can have zero or more labels. A block has a body that enables you to declare arguments and nested blocks.
- Arguments are used to assign a value to a name.
- An expression represents a value that can be assigned to an identifier.

Proprietary + Confidential

Terraform can be used on multiple public and private clouds

- Considered a first-class tool in Google Cloud
- Already installed in Cloud Shell

```
provider “google” {
   region      = “us-central1”
}

resource “google_compute_instance” {
   name        = “instance name”
   machine_type= “n1-standard-1”
   zone        = “us-central1-f”

   disk {
      image = “image to build instance”
   }
}

output “instance_ip” {
   value = “${google_compute.ip_address}”
}
```

Google Cloud

Terraform can be used on multiple public and private clouds. Terraform is already installed in Cloud Shell.

The example Terraform configuration file shown starts with a provider block that indicates that Google Cloud is the provider. The region for the deployment is specified inside the provider block.

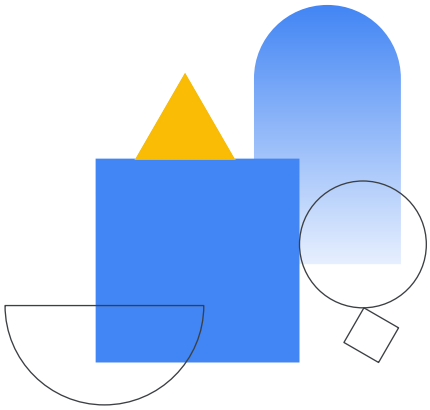
The resource block specifies a Google Cloud Compute Engine instance, or virtual machine. The details of the instance to be created are specified inside the resource block.

The output block specifies an output variable for the Terraform module. In this case, a value will be assigned to the output variable "instance_ip."

Proprietary + Confidential

Lab Intro

Building a DevOps Pipeline



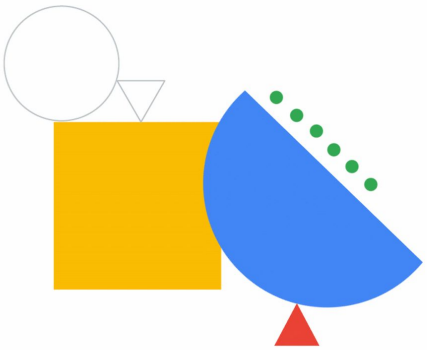
Google Cloud

In this first lab, you will build a DevOps pipeline using Cloud Source Repositories, Cloud Build, and Artifact Registry. Specifically, you will create a Git repository. You will then write a simple Python application and add it to your repository. After that you will test your web application in Cloud Shell and then define a Docker build.

Once you define the build, you will use Cloud Build to create a Docker image and then store the image in Artifact Registry. Then, you will see how to automate builds using triggers.Once you have a trigger, you will test it by making a change to your program and pushing that change to your Git repo.

Proprietary + Confidential

Review:
DevOps Automation



In this module, you learned about services that you can use to help automate the deployment of your cloud resources and services. You used Cloud Source Repositories, Cloud Build, triggers, and Artifact Registry to create continuous integration pipelines. A continuous integration pipeline automates the creation of deployment packages like Docker images in response to changes in your source code.

You also saw how to automate the creation of infrastructure using the infrastructure as code tool, Terraform.